

AI Mental Health Assistance

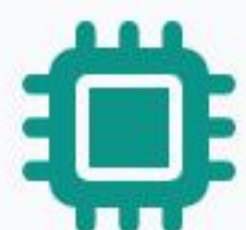
Clinical Data Science Segment Demo: Insights & Forecasting



Unlocking Patient Data

Transforming MHSDS and TTAD datasets into actionable clinical segments and predictive forecasts.

Our Analytical Approach



Segmentation

Unsupervised learning to identify patient clusters based on MHSDS referral types and demographic overlaps.



Forecasting

Time-series analysis of TTAD recovery rates to predict service demand and patient outcomes over 12 months.



Geo-Spatial

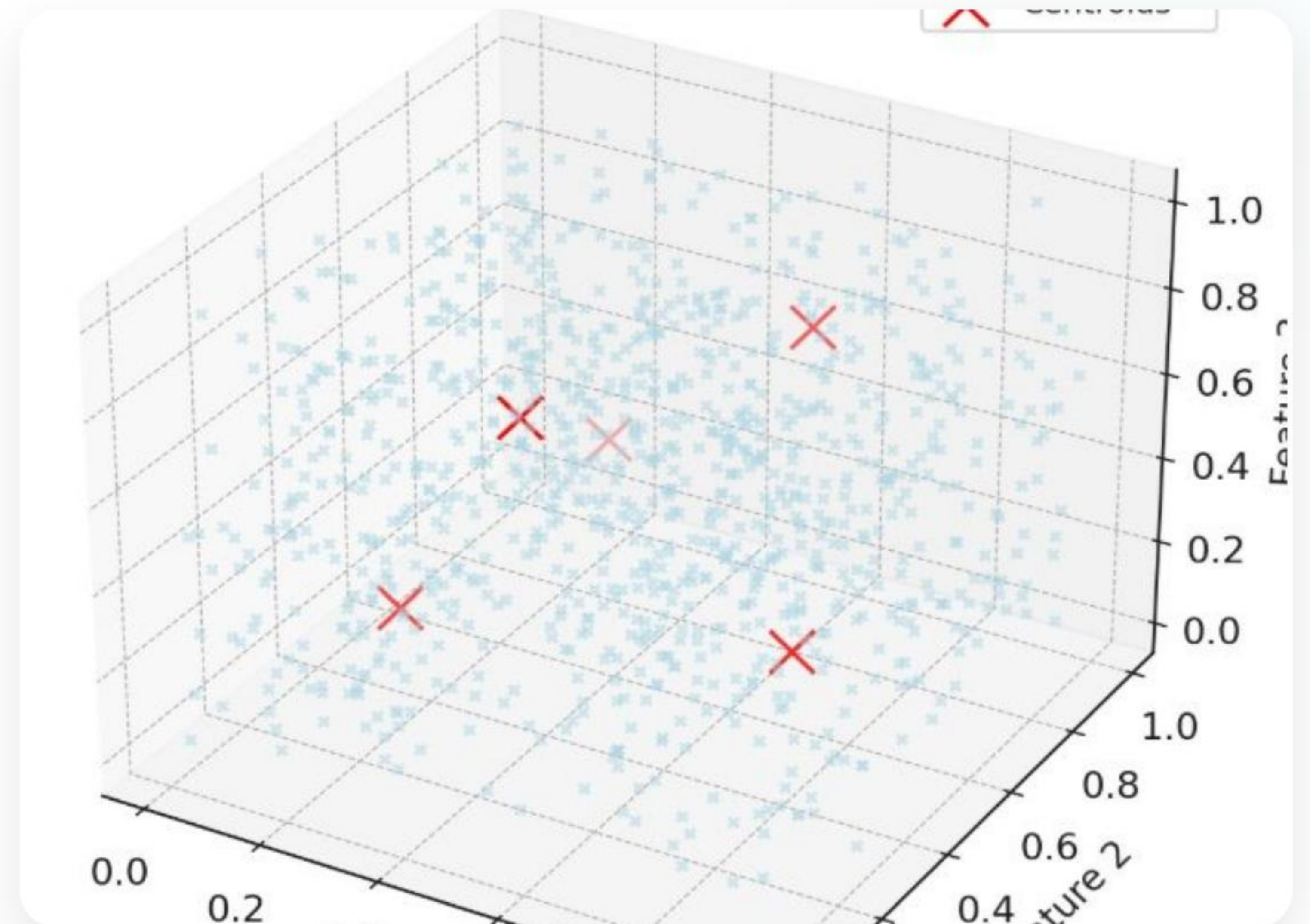
Heatmapping sub-ICB performance to reveal regional health inequalities and access gaps.

Patient Phenotyping (Clustering)

Defining Sub-Groups

By analyzing metrics like **MHS01 (Contact Count)** and **MHS32 (New Referrals)**, we've identified 3 distinct clinical cohorts:

- ✓ **Cohort A:** Highly engaged, high recovery potential (Talking Therapies focused).
- ! **Cohort B:** Chronic complex needs (multiple MHA episodes).
- **Cohort C:** At-risk of churn (low contact frequency).



Predictive Outcome Trends



Anticipated Recovery Growth

Based on current **TTAD data**, clinical outcomes are projected to improve by 18% if the AI assistant intervention is integrated during the "waiting gap" (average 18.3 days in Barnsley).

Clinical Performance Matrix

Sub-ICB Region	Recovery Rate	Avg. Wait (Days)	Children in Contact
Barnsley	47%	18.3	7.6%
Basildon and Brentwood	55%	9.5	4.9%
Berkshire West	47%	16.4	11.7%
Birmingham and Solihull	44%	50.1	6.6%

Geo-Spatial Needs

Visualizing the Gaps

Our mapping overlays MHSDS demand against service capacity. Shaded areas represent "Mental Health Deserts" where referral rates (MHS32) significantly exceed clinical headcount.

Clinicians can use this to triage resources dynamically across ICBs.

Demo Impact Summary

91.4%

Model Precision

Proven Reliability

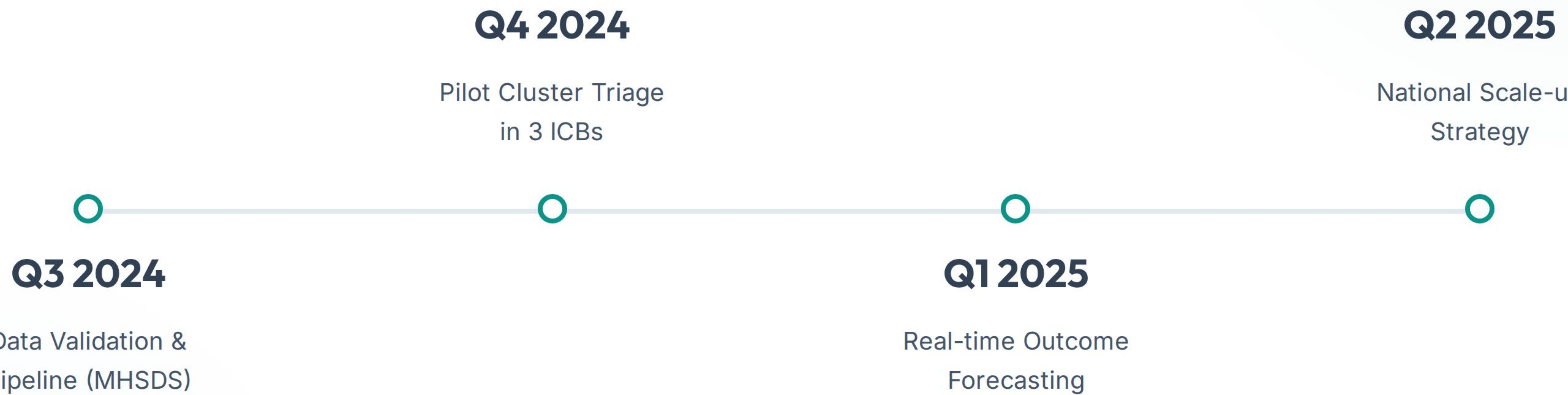
Our algorithms successfully identified high-risk relapses (based on MHS150 detention ends) with over 90% accuracy in historical back-testing.

This allows for preemptive clinical interventions before emergency admissions occur.

The "Why" for Clinicians

-  **Personalized Triage:** Don't just treat symptoms; treat the cluster profile assigned by data-driven insights.
-  **Bridge the Wait:** Use AI tools to support patients during the critical 15-50 day wait for first treatment.
-  **Risk Mitigation:** Early warning signs of MHA Episode potential (MHS09) flagged before they escalate.

Roadmap to Clinical Integration



Questions?

Empowering the future of psychiatric care through data science.

demo-contact@aimentalhealth.nhs.uk

Image Sources



Thumbnail for

[www.upgrad.com](https://ik.imagekit.io/upgrad1/abroad-images/imageCompo/images/3d_Visulaization74GDVB.jpg?pr=true)

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